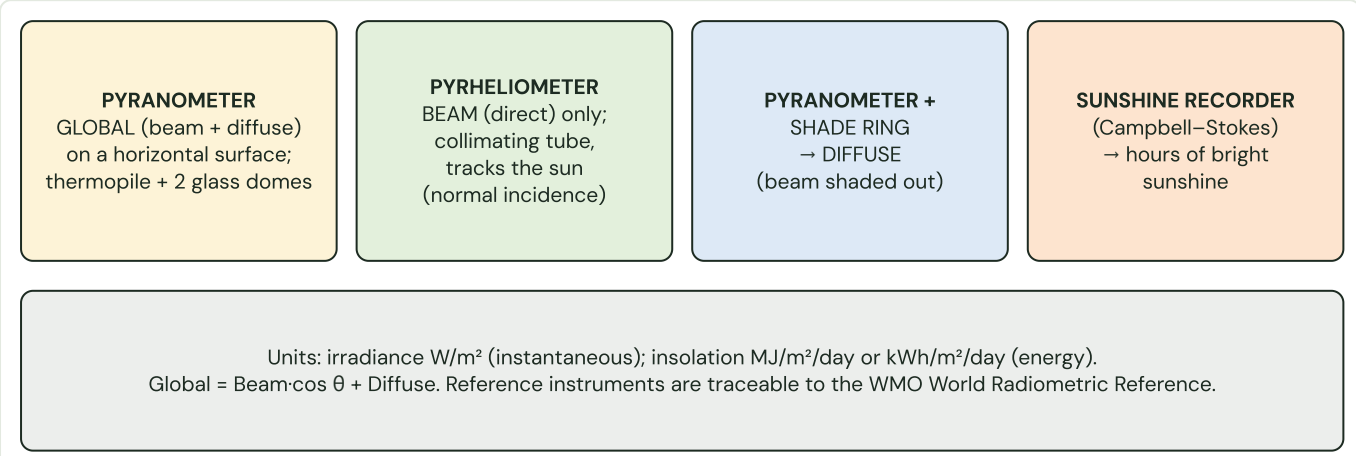
UNIT 7 · TOPIC 2(B)

Measurement of Solar Radiation

Syllabus topic: Solar radiation: measurement and characteristics of the solar spectrum



Measurement of solar radiation (redrawn from the source figure)

Instrument	Measures	How
PYRANOMETER	GLOBAL (beam + diffuse) radiation on a horizontal surface	Black-and-white or thermopile sensor under two glass domes; voltage ∝ irradiance (W/m²)
PYRHELIOMETER	BEAM (direct) radiation at NORMAL INCIDENCE	Collimating tube that tracks the sun
Pyranometer + SHADE RING / shading disc	DIFFUSE radiation	The ring blocks the beam
SUNSHINE RECORDER (Campbell–Stokes)	HOURS of bright sunshine (not intensity)	Burns a trace on a card

Quantities and units

Quantity	Unit
IRRADIANCE (instantaneous power)	W/m²
INSOLATION (energy received)	MJ/m²·day or kWh/m²·day

$$\text{Global} = \text{Beam}\cdot\cos\theta + \text{Diffuse}$$

- Reference instruments are traceable to the **WMO World Radiometric Reference**.
- Solarimeters/actinographs and silicon reference cells are also used.
- The data feed collector and PV-system sizing (Topics 3–4).

Practice MCQs

Q22. A pyranometer measures:

(a) Beam only
(b) Global (beam + diffuse) radiation
(c) Sunshine hours
(d) Wind speed

Q23. A pyrheliometer measures:

(a) Diffuse radiation
(b) Beam (direct) radiation at normal incidence
(c) Global
(d) Temperature

Q24. To measure DIFFUSE radiation, a pyranometer is fitted with a:

(a) Second dome
(b) Shade ring/disc to block the beam
(c) Lens
(d) Tracker

Q25. A pyrheliometer must:

(a) Lie flat
(b) Track the sun (point at normal incidence)
(c) Be shaded
(d) Face north

Q26. Hours of bright sunshine are recorded by a:

(a) Pyranometer
(b) Campbell–Stokes sunshine recorder
(c) Anemometer
(d) Barometer

Q27. Irradiance is expressed in:

(a) W/m²
(b) MJ/day
(c) °C
(d) m/s

Q28. Daily insolation (energy) is expressed in:

(a) W/m²
(b) MJ/m²/day or kWh/m²/day
(c) N/m²
(d) kg

Q29. Typical daily global insolation over most of India is about:

(a) 1–2
(b) 5–7
(c) 15–20
(d) 30 kWh/m²/day

Q35. Reference solar instruments are traceable to the:

(a) WMO World Radiometric Reference
(b) NIST clock
(c) ISO 9001
(d) OECD code

Answers

Q22 — (b) Global (beam + diffuse) radiation — *Pyranometer = global radiation.*
Q23 — (b) Beam (direct) radiation at normal incidence — *Pyrheliometer = beam (sun-tracking).*
Q24 — (b) Shade ring/disc to block the beam — *Pyranometer + shade ring = diffuse.*
Q25 — (b) Track the sun (point at normal incidence)
Q26 — (b) Campbell–Stokes sunshine recorder — *Campbell–Stokes = sunshine hours.*
Q27 — (a) W/m²
Q28 — (b) MJ/m²/day or kWh/m²/day
Q29 — (b) 5–7
Q35 — (a) WMO World Radiometric Reference

Sources: Sukhatme & Nayak — Solar Energy: Principles of Thermal Collection and Storage; Rai — Non-Conventional Energy Sources; Duffie & Beckman — Solar Engineering of Thermal Processes; WMO Guide to Meteorological Instruments (radiation); MNRE/PIB releases (capacity figures confirmed against MNRE data to 30 June 2026); NPTEL renewable-energy lectures.